

A Hybrid Smishing Detection Framework: Integrating Machine Learning, Rule-Based Filtering, and Domain Reputation Analysis

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Abstract—The rise of SMS phishing (smishing) attacks poses a significant threat to mobile security, particularly for Vietnamese users whose linguistic environment is inadequately served by existing international solutions. This paper makes three primary contributions. First, we construct a Vietnamese smishing dataset comprising 2,600 labeled SMS messages, collected from real-world devices and community-sourced reports. Second, we propose a Hybrid Smishing Detection Framework featuring a four-layer linguistic preprocessing pipeline designed to handle Vietnamese text obfuscation, including teencode, Leetspeak, and broken Telex input. Third, we integrate this pipeline with a tree-based ML classifier and a Hybrid Decision Logic that incorporates Domain Verification. On a held-out test set of 520 samples, the XGBoost-based system achieves an overall accuracy of 97.5% and a smishing recall of 78%, outperforming single-model baselines in the precision-recall trade-off relevant to cyber-security deployment.

Index Terms—Smishing, SMS Phishing, Machine Learning, Vietnamese NLP, Linguistic Feature Extraction, Hybrid Architecture

I. INTRODUCTION

In the modern digital era, SMS remains one of the most reliable communication channels, with an open rate exceeding 98% [1]. This trust has made it a prime vector for *smishing* (SMS Phishing)—a form of social engineering where attackers deceive victims into revealing sensitive information or installing malware. In Vietnam, attackers routinely impersonate banks, e-commerce platforms, and government agencies, often exploiting Brandname abuse to bypass traditional filters, causing significant financial and data losses for unsuspecting users.

Existing defenses remain limited: static blacklists and simple keyword matching fail against obfuscated text (teencode, Leetspeak, mixed-character URLs) and dynamic URL shorteners. Advanced ML pipelines from the international literature are furthermore optimized for English; applying them to Vietnamese yields suboptimal results due to diacritical marks, tone-dependent semantics, and local obfuscation conventions. To the best of our knowledge, no comprehensive system currently adapts these architectures specifically for the Vietnamese linguistic and telecommunication landscape.

A. Task Definition

We define smishing detection as a binary text classification problem. Given an incoming SMS message T with sender type S , the system must assign a label $L \in \{\textit{Smishing}, \textit{Legitimate}\}$. Formally:

- **Input:** A tuple (T, S) , where T is the raw SMS body and S denotes the sender type (e.g., Brandname, Phone Number, Unknown).
- **Output:** A label L , a confidence score $\text{Score} \in [0, 1]$, and a natural-language explanation.

Smishing example: “[VCB] TK của bạn đã bị KHOA do đang nhập sai nhiều lần. Vui lòng truy cập <http://vcb-xacnhan.xyz> để mở khoa ngay.” — domain spoofing (`vcb-xacnhan.xyz` impersonating Vietcombank) combined with urgency cues are two hallmarks of smishing.

Legitimate example: “Anh ơi anh về chưa? Em đang doi o nha. Nho mua com nhe.” — a personal conversational message with first-person pronouns and everyday activities, no URLs, and no financial pressure.

B. Contributions

To address the challenges above, we propose a Hybrid Real-Time Smishing Detection System tailored for Vietnamese. Our key contributions are:

- 1) **Dataset:** A composite Vietnamese dataset of 2,600 labeled SMS samples built via dual-channel acquisition and a five-annotator majority-vote protocol.
- 2) **Framework:** A four-layer linguistic preprocessing pipeline for Vietnamese, combined with an ML classifier and deterministic Domain Verification integrated through a Hybrid Decision Logic.
- 3) **Experiments:** Comparative evaluation of Decision Tree, Random Forest, and XGBoost demonstrating our hybrid approach outperforms TF-IDF baselines in the precision-recall trade-off critical for cyber-security deployment.

II. RELATED WORKS

A. Smishing Detection Architectures

The task of automatically detecting smishing remains nascent compared to email spam detection, primarily due to short message length and data scarcity. Mishra and Soni [1] present DSmsishSMS, a Defense-in-Depth framework coupling a Domain Checking Phase (URL validation via search-engine signatures) with an SMS Classification Phase based on five hand-crafted features: misspelled words, symbols, special characters, smishing keywords, and Leet words. Their Backpropagation Algorithm achieves 97.93% accuracy, outperforming Random Forest. Our work adopts a similar hybrid philosophy but extends it to Vietnamese with a richer, language-specific feature pipeline.

B. The Challenge of Leet Obfuscation

Mishra and Soni found that 74.22% of smishing messages contain character substitutions (e.g., “P@ym” → “Paytm”) to bypass keyword filters [1]. Li and Zeng [2] further document how such substitutions systematically evade dictionary-based detectors. In Vietnamese smishing, Leetspeak interacts with tone marks and Telex input artifacts, creating a more complex obfuscation landscape than prior English-focused studies address.

C. Text Normalization for SMS Spam Detection

Goel *et al.* [5] propose an ML-driven smishing detection framework that applies text normalization as a pre-processing step to convert informal SMS language—abbreviations, slang, and short forms—into a standardized form before classification. Using a Naïve Bayes classifier, they achieve 96.2% accuracy with a false-positive rate of 3.87% on an English dataset. Our work extends this normalization philosophy to the substantially more complex Vietnamese setting, where tone-mark interactions, Telex input artifacts, and brandname spoofing demand a multi-layer, language-specific pipeline rather than a single normalization pass.

D. SMS Spam Datasets and the Case for Language-Specific Corpora

Almeida *et al.* [4] introduced the widely-used UCI SMS Spam Collection, comprising 5,574 English-language messages, which became the de facto benchmark for SMS spam research. While instrumental in driving English-centric progress, this dataset does not capture the linguistic properties of other languages. Our work addresses this gap by constructing a Vietnamese-specific corpus of 2,600 labeled SMS messages through a dual-channel acquisition strategy, enabling evaluation under Vietnamese obfuscation conventions and Brandname-abuse patterns that no existing public dataset covers.

E. Tree-Based Ensemble Classifiers in Cybersecurity

Gradient-boosted decision trees have demonstrated consistent superiority over single-tree and linear models on tabular feature sets. Chen and Guestrin [3] introduced XGBoost, a scalable end-to-end tree boosting system that incorporates a sparsity-aware split-finding algorithm and second-order gradient statistics, achieving state-of-the-art results across numerous structured-data competition benchmarks. In the smishing context, a 28-dimensional dense feature vector derived from our linguistic pipeline constitutes exactly the kind of structured, heterogeneous input for which gradient boosted trees excel over both shallow models (Decision Tree) and sparse text representations (TF-IDF).

III. DATA COLLECTION

Due to the scarcity of standardized Vietnamese smishing datasets, we constructed a composite corpus through a dual-channel acquisition strategy.

A. Manual Transcription from Public Sources

The first channel harvested confirmed smishing messages frequently reported as screenshots on Vietnamese cybersecurity forums and anti-scam community groups. We performed manual transcription rather than OCR to accurately preserve linguistic nuances—teencode, special character obfuscations, and malicious URLs—that automated recognition tools often corrupt. This yielded a high-quality set of distinct smishing signatures targeting banking and e-commerce sectors.

B. Real-World Data Acquisition via Mobile Utility

The second channel collected real-world SMS data—legitimate conversations, OTPs, and carrier advertisements—from real Android devices using *SMS Backup & Restore*, with explicit consent from participants. This application exports local message repositories into a structured XML format, allowing efficient extraction of message body, timestamp, and sender information.

C. Dataset Summary and Annotation Protocol

Following aggregation, cleaning, and normalization, the final dataset comprises 2,600 unique rows (Table I). Labels were assigned by five annotators using pre-defined guidelines: a message is *Smishing* if it contains (1) a URL leading to a non-whitelisted or suspicious domain, (2) impersonation of a trusted entity, (3) financial pressure or urgency combined with a request for personal information, or (4) unrealistic recruitment offers. Final labels were decided by majority vote; unresolved samples were discarded to maintain dataset purity.

TABLE I
DATASET CLASS DISTRIBUTION

Class	Count	Percentage
Legitimate (Ham)	2,353	90.5%
Smishing (Spam)	247	9.5%
Total	2,600	100%

IV. DATA PREPROCESSING

SMS data contains significant noise and lacks structural uniformity. In spam messages, obfuscation techniques—inserting random spaces within URL segments, applying Leetspeak, or mixing characters—actively bypass keyword-based filters. A single-pass normalization approach fails to address these interacting noise sources, motivating our four-layer design. Each layer targets a distinct noise category; their outputs also serve as discriminative classifier features.

A. Layer 1 — Entity Masking

Specific entities are replaced with standardized tokens (e.g., <URL>, <PHONE>, <MONEY>) using an `OrderedDict` to enforce conflict-free processing order. The pipeline executes in the following order:

- **Platform Links:** Zalo and Telegram links (<LINK>).
- **Emails, URLs:** Standard email regex; custom logic for obfuscated malicious links (<EMAIL>, <URL>).

- **Bank Accounts, Telecommunications:** Context-aware extraction for hotlines, landlines, and mobile numbers.
- **Datetime, Monetary Values, Codes & OTPs:** <TIME>, <MONEY>, <CODE>.
- **Anti-Evasion URL Detection:** Detects intentional spacing in domains (e.g., bit.ly/abc), and protocol-agnostic domains matching high-risk TLD patterns (.xyz, .top, .moe).

B. Layer 2 — Text Normalization and Leetspeak Decoding

The TextNormalizer standardizes the character set and decodes obfuscations via a weighted scoring system validated against a Vietnamese dictionary. Three obfuscation levels are recognized:

- **Symbol Leet (weight 1.2–1.5):** e.g., @→a, !→i. Highly indicative of spam.
- **Visual Leet (weight 1.0):** e.g., 0→o, 4→a (h0tro → htro).
- **Teencode (weight 0.3):** e.g., ko→khong.

A placeholder mechanism protects Layer 1 tokens from corruption during separator removal.

C. Layer 3 — Whitelist Filtering

Valid out-of-dictionary tokens—brand names (Shopee, VCB), technical jargon (OTP, 4G), and SMS abbreviations (lh, sdt)—are exempted from further processing to prevent erroneous spell-check corrections (e.g., “VCB” → “viec”).

D. Layer 4 — Spell Check and Anomaly Detection

Remaining Out-Of-Vocabulary (OOV) tokens are analyzed by a Dual-Lookup Mechanism (full-diacritics dictionary + accent-free shadow dictionary). Tokens failing both checks are tagged as one of four error categories used as classifier features: *Telex Artifacts* (aa, aw, dd), *Run-On Words* (bisection heuristic, e.g., vui + long), *Gibberish Strings* (violating Vietnamese phonotactics), or *Character Repetition* (3+ identical characters, e.g., HOTTT). A Data Capacity Whitelist proactively exempts telco tokens (e.g., 3gb, 12mbps) that are technically OOV but valid in SMS contexts.

V. PROPOSED SYSTEM METHODOLOGY

The framework combines a Machine Learning scorer, a Domain Verification module, and a Hybrid Decision Logic. A purely statistical model is vulnerable to adversarial evasion; a purely rule-based system cannot generalize to novel attack patterns. The hybrid architecture addresses both shortcomings.

A. Machine Learning Model

The raw SMS T is passed through the 4-Layer Pipeline (Section IV) to generate a 27-dimensional linguistic vector F_{vec} . Unlike sparse TF-IDF matrices, this dense vector directly encodes obfuscation indicators prevalent in Vietnamese fraud. The Sender Type S (Brandname, Phone Number, etc.) is label-encoded and appended:

$$V = \text{COMBINE}(F_{\text{vec}}, S) \in \mathbb{R}^{28} \quad (1)$$

The pre-trained XGBoost model M_{AI} produces a probability score:

$$P_{AI} = M_{AI}.\text{predict_proba}(V), \quad P_{AI} \in [0, 1] \quad (2)$$

A score near 1 reflects high obfuscation and urgency cues; near 0 indicates legitimate patterns. P_{AI} feeds the Hybrid Decision Logic.

B. Domain Verification

The Domain Verification module provides deterministic, AI-independent verdicts for URL-bearing messages (Fig. 1). The brand context B and all URLs are extracted; tldextract isolates the registered domain D to prevent subdomain spoofing (e.g., distinguishing vietcombank.ngrok.io as belonging to ngrok.io). If $D \in W_{\text{static}}$ (a pre-compiled whitelist of major Vietnamese banks, telcos, and global platforms), the domain is flagged Legitimate (Risk=−1.0). Otherwise, two-level dynamic verification triggers: Level 1 checks whether D appears in the top-5 search results for the brand name; Level 2 scans the site’s source for internal links pointing to verified domains. An unreachable, redirect-heavy, or unverifiable site yields Risk=1.0.

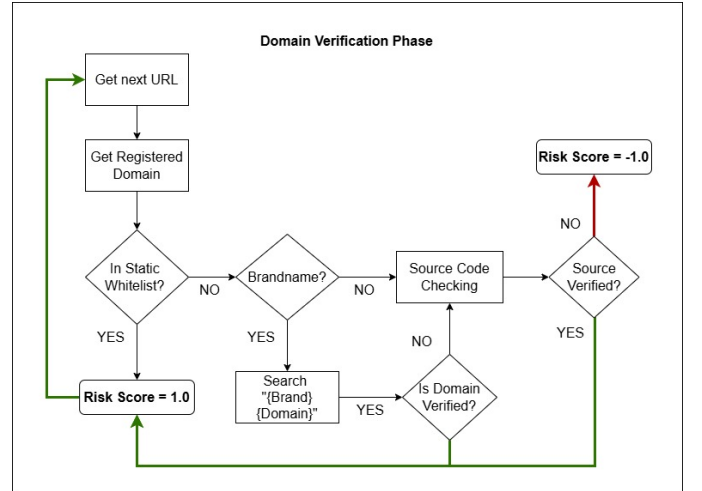


Fig. 1. Architecture of the Domain Verification Phase.

C. Hybrid Decision Logic

The Hybrid Decision Logic applies a three-tier waterfall (Algorithm 1; system architecture in Fig. 2).

Tier 1 — Domain-Driven Verdict: If $\text{Risk} \geq 0.8$, the message is immediately classified as Smishing. For trusted domains ($\text{Risk} = -1.0$), user-generated-content platforms undergo a secondary AI check ($P_{AI} > 0.65$).

Tier 2 — Context-Aware AI Verdict: When no URL is present or domain status is inconclusive, P_{AI} is evaluated alongside context flags. Conversational messages override high AI risk; messages with danger keywords from unknown senders force a Smishing verdict.

Tier 3 — Explainable Output: Every decision yields a tuple (Label, Score, Decision Phase, Reasoning) for user and analyst transparency.

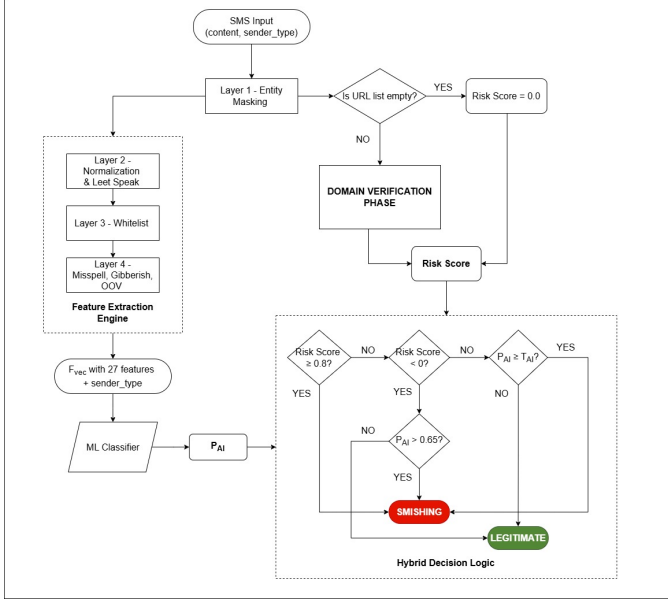


Fig. 2. Architecture of the proposed Hybrid Smishing Detection System.

Algorithm 1 Hybrid Smishing Detection Algorithm

Require: Incoming SMS T , Sender Type S , Thresholds T_{AI} ,

T_{high}

Ensure: Label (*Smishing* / *Legitimate*)

- 1: $F_{vec} \leftarrow \text{ExtractFeatures}(T)$ {Layers 1–4}
- 2: $V \leftarrow \text{Combine}(F_{vec}, S)$
- 3: $P_{AI} \leftarrow M_{AI}.\text{predict_proba}(V)$
- 4: $Risk \leftarrow 0$
- 5: **if** T contains URL **then**
- 6: $Risk \leftarrow \text{VerifyDomain}(T)$
- 7: **end if**
- 8: **if** $Risk \geq T_{high}$ **then**
- 9: **return** *Smishing* {Tier 1: High-Risk Domain}
- 10: **else if** $Risk = -1.0$ **then**
- 11: **if** $P_{AI} > 0.65$ **then**
- 12: **return** *Smishing (Warning)* {UGC Exception}
- 13: **else**
- 14: **return** *Legitimate*
- 15: **end if**
- 16: **else**
- 17: **if** $P_{AI} \geq T_{AI}$ **then**
- 18: **return** *Smishing*
- 19: **else**
- 20: **return** *Legitimate*
- 21: **end if**
- 22: **end if**

VI. EVALUATION AND RESULTS

All Precision, Recall, and F1 values refer to the *Smishing* minority class, the decision-critical class in cyber-security deployment. Metrics are reported on $[0, 1]$.

A. Experimental Setup

The 2,600-message dataset (90% Legitimate, 10% Smishing) was split 80/20 using stratified sampling ($\text{random_state}=42$), yielding a test set of 520 messages (465 Legitimate, 55 Smishing). We adopt a two-level strategy: (i) three standalone classifiers evaluated on the full 520-sample test set using 28-dimensional features, and (ii) the complete Hybrid System evaluated on 100 stratified messages with per-error categorization.

Rather than a fixed threshold of 0.5, each classifier’s optimal threshold τ^* is selected as the maximum-F1 operating point on the Precision-Recall curve:

$$\tau^* = \arg \max_{\tau} \frac{2 \cdot P(\tau) \cdot R(\tau)}{P(\tau) + R(\tau)}.$$

The Smishing F1-Score is the primary comparison metric because overall Accuracy is misleading under a 90/10 class imbalance.

B. Feature Engineering versus TF-IDF Baseline

We compare the 4-Layer Pipeline against a TF-IDF Baseline (5,000 unigrams/bigrams augmented with `has_url`, `has_phone`, `sender_type`; fixed threshold 0.5). Results are in Table II.

TABLE II
TF-IDF BASELINE VS. PROPOSED 4-LAYER LINGUISTIC PIPELINE

Feature Set	Model	Acc.	Prec.	Recall	F1	ROC-AUC
TF-IDF Baseline ($\tau=0.5$)	Decision Tree	0.9231	0.6000	0.8182	0.6923	0.8831
	Random Forest	0.9673	0.9750	0.7091	0.8211	0.9844
	XGBoost	0.9673	0.8800	0.8000	0.8381	0.9696
4-Layer Linguistic ($\tau=\tau^*$)	Decision Tree	0.9519	0.7679	0.7818	0.7748	0.9050
	Random Forest	0.9654	0.9744	0.6909	0.8085	0.9730
	XGBoost	0.9750	0.9773	0.7818	0.8687	0.9659

Prec., Recall, F1 are for the Smishing class.

The linguistic pipeline yields the most pronounced gains for XGBoost: Smishing F1 rises from 0.838 to **0.869** (+3.1 pp), and—crucially for operational deployment—Precision rises from 0.88 to **0.977** (+9.7 pp), meaning fewer than 2.3% of flagged messages are false alarms. Decision Tree benefits most in F1 (+8.3 pp), confirming that hand-crafted obfuscation indicators carry discriminative signal that sparse TF-IDF cannot capture for leet-encoded Vietnamese text.

C. Standalone Model Results and Analysis

Table III reports full results at optimal thresholds.

Decision Tree: The Decision Tree detects **43 of 55** smishing messages (Recall = 0.78), but at the cost of **13 false positives**—roughly 1 in 4 flagged messages is actually benign (Precision = 0.768), an unacceptable disruption rate in production. Overall accuracy is 95.2%, but this masks the high false-alarm rate.

TABLE III
SMISHING-CLASS METRICS AT OPTIMAL THRESHOLD (4-LAYER
LINGUISTIC FEATURES; TEST SET $n=520$)

Model	τ^*	Acc.	Prec.	Recall	F1	TP	FP
Decision Tree	0.8942	0.9519	0.7679	0.7818	0.7748	43	13
Random Forest	0.6842	0.9654	0.9744	0.6909	0.8085	38	1
XGBoost	0.8336	0.9750	0.9773	0.7818	0.8687	43	1

FN = 55 - TP; TN = 465 - FP.

Random Forest: Random Forest produces only **1 false positive** across 465 legitimate messages (Precision = 0.974), but achieves this by missing **17 smishing attacks** (Recall = 0.69; 31% miss rate). In a security context, almost one in three attacks reaching the user undetected is not acceptable. Random Forest suits scenarios where false alarms carry higher cost than missed attacks.

XGBoost: XGBoost achieves the best trade-off for cyber-security deployment. It detects the same **43 of 55** smishing messages as Decision Tree while emitting only **1 false positive**—catching **5 more attacks** than Random Forest (+8.3 pp Recall) at equal false-alarm cost. Its Smishing F1 of **0.8687** is the highest of the three, and its overall Accuracy of **0.9750** reflects strong performance on both classes. XGBoost is therefore selected as the core classifier of the Hybrid Detection Framework.

D. Error Analysis of the Full Hybrid System

The complete Hybrid System ($\tau \approx 0.46$) was applied to **100 stratified messages**. Of these, **87 were correctly classified**, yielding **3 False Negatives (FN)** and **10 False Positives (FP)**, summarized in Table IV.

TABLE IV
ERROR ANALYSIS ON 100 SAMPLED MESSAGES (HYBRID SYSTEM)

Type	Category	Count
FN	No-URL / low-obfuscation smishing below AI threshold	3
FP	AI model over-triggered on transactional messages	5
FP	Legitimate domain absent from Domain Risk whitelist	5
Total errors		13

False Negatives: All three FNs are smishing messages where neither the AI model nor the Domain Risk module exceeded their respective thresholds. Three distinct failure patterns were identified:

- 1) **Authority impersonation without a URL.** A message impersonating *Cảnh sát Giao thông*—containing only a hotline number—received $P_{AI}=0.20$ because it is written in clean, formal Vietnamese that lacks the obfuscation cues the model relies on. The Domain Risk module had no URL to evaluate.
- 2) **Telegram social-engineering link with minimal obfuscation.** A message linking to `t.me/hn_silent` was scored Risk=0.0 because `t.me` is whitelisted as a known messaging platform, and the AI gave $P_{AI}=0.09$

as the text resembles a social invitation rather than a financial scam.

- 3) **Leet-speak phishing URL just below the decision boundary.** A VNeID impersonation with leet-encoded text obtained $P_{AI}=0.45$ —one point below $\tau^*\approx 0.46$ —while the Domain Risk module returned $r=0.0$ for `vneid-update.t0p`, finding no conclusive blacklist evidence.

These cases motivate: (a) a recall-biased rule for URL-free government/law-enforcement impersonation, and (b) a pattern-based leet-URL detector for domains whose structure matches known brand names.

False Positives: AI over-triggered on transactional messages (5 cases). Five legitimate confirmation messages (restaurant reservations, taxi bookings, e-commerce notifications) were flagged because they share structural features with smishing: brevity, a specific time reference, and an entity name (e.g., a reservation confirmation scored $P_{AI}=0.94$). The model has learned “*short message + time + entity*” as a risk pattern without distinguishing transactional confirmations from fraud.

Legitimate domain absent from the whitelist (5 cases). Five messages were escalated by the Domain Risk module because their domains were absent from the static whitelist and the live-search verifier returned inconclusive results. Affected domains include `thuedientu.gdt.gov.vn` (official tax portal), `ghn.vn` (GHN Express), and `shopeepay.vn`. Concrete remedies: extend the whitelist to all `*.gov.vn` and `*.edu.vn` subdomains, and apply a low-risk prior for government and academic TLDs in the live-search verifier.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a Hybrid Smishing Detection Framework purpose-built for Vietnamese, combining a four-layer linguistic preprocessing pipeline with an XGBoost classifier and a deterministic Domain Verification engine. The pipeline converts raw, obfuscated Vietnamese SMS into a dense 28-dimensional feature vector, raising Smishing F1 from 0.838 to **0.869** and Precision from 0.88 to **0.977** over a TF-IDF baseline. The three-tier Hybrid Decision Logic achieves **97.5%** accuracy and **78.2%** Smishing Recall on the held-out test set, and correctly classifies **87%** of real-world samples in the full-pipeline evaluation. Residual errors concentrate in two identifiable failure modes—URL-free authority-impersonation and whitelist gaps—that directly inform the next development iteration.

B. Limitations and Future Work

Dataset scale. At 2,600 samples with only 247 positive examples, the corpus provides limited exposure to rare attack subtypes such as crypto investment fraud or OTP relay attacks.

Static rule maintenance. The hand-crafted pipeline rules require manual updates as attackers adapt; both the whitelist and the Domain Verification module demonstrated blind spots in the error analysis.

URL dependency. All three FNs involved messages without a suspicious URL. When no actionable URL is present, the system relies entirely on the 28-dimensional feature vector, which does not capture semantic intent.

Future directions: (1) *LLM-Driven Dataset Expansion* to 10,000+ samples (GPT-4/Claude/Gemini generating hard negatives and adversarial smishing with rationale annotations); (2) *Neural Text Normalization* replacing regex rules with a BARTpho-based Seq2Seq normalizer trained on ViLexNorm [6]; and (3) *Rationale-Aware Knowledge Distillation* training a compact ViSoBERT [7] student via $\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{Hard}} + \beta\mathcal{L}_{\text{Soft}} + \gamma\mathcal{L}_{\text{Rationale}}$ to capture semantic intent in URL-free attacks and suppress transactional false positives.

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